BMC Helix Technical Bootcamp - ITSM/ITOM Container

Lab 4: Setting Pre-requisites to Deploy BMC Helix Service Management applications

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OBJECTIVE: In this lab, you will check that Yum is fully up to date and verify or install other required items before commencing with the ITSM installation in the next lab.

Refer to the full product documentation for complete implementation details. The steps provided in this lab document are simplified and tailored to our specific environment. It is recommended that you review the full documentation before beginning any lab work: https://docs.bmc.com/xwiki/bin/view/Service-Management/On-Premises-Deployment/BMC-Helix-Service-Management-Deployment/brid25101/Installing/Preparing-for-installation/Setting-up-BMC-Deployment-Engine/

Task 1: Update Yum

In this task you will run commands as a user with sudo access to verify Yum is installed, configured and working. All commands will be executed as root in a deploymentengine session.

1. Due to a known issue with the Postgres version in the training environment, we need to remove the Postgres packages before updating yum. Log into **deploymentengine** with credentials **root**/**password** and run the following command:

```
dnf remove postgres*
```

2. Confirm removal of packages by typing **y**.

```
[root@deploymentengine ~]# dnf remove postgres*
Dependencies resolved.
================================================================================================
 Package                                Arch        Version                      Repository       Size
================================================================================================
Removing:
 postgresql                             x86_64      13.22-1.el9_6                @appstream      5.9 M
 postgresql-private-libs                x86_64      13.22-1.el9_6                @appstream      335 k
 postgresql15                           x86_64      15.14-1PGDG.rhel9            @pgdg15         8.4 M
 postgresql15-contrib                   x86_64      15.14-1PGDG.rhel9            @pgdg15         2.7 M
 postgresql15-devel                     x86_64      15.14-1PGDG.rhel9            @pgdg15          18 M
 postgresql15-libs                      x86_64      15.14-1PGDG.rhel9            @pgdg15         1.0 M
 postgresql15-server                    x86_64      15.14-1PGDG.rhel9            @pgdg15          25 M
Removing unused dependencies:
 clang                                  x86_64      19.1.7-1.el9                 @appstream      181 k
 clang-devel                            x86_64      19.1.7-1.el9                 @appstream       25 M
 clang-libs                             x86_64      19.1.7-1.el9                 @appstream      413 M
 clang-resource-filesystem              x86_64      19.1.7-1.el9                 @appstream       15 k
 clang-tools-extra                      x86_64      19.1.7-1.el9                 @appstream       69 M
 cmake-filesystem                       x86_64      3.26.5-2.el9                 @appstream        0
 compiler-rt                            x86_64      19.1.7-1.el9                 @appstream       37 M
 gcc-toolset-14-gcc                     x86_64      14.2.1-7.1.el9               @appstream      122 M
 gcc-toolset-14-gcc-c++                 x86_64      14.2.1-7.1.el9               @appstream       39 M
 gcc-toolset-14-libstdc++-devel         x86_64      14.2.1-7.1.el9               @appstream       22 M
 libedit-devel                          x86_64      3.1-38.20210216cvs.el9       @appstream       57 k
 libicu-devel                           x86_64      67.1-10.el9_6                @appstream      5.4 M
 libomp                                 x86_64      19.1.7-1.el9                 @appstream      1.9 M
 libomp-devel                           x86_64      19.1.7-1.el9                 @appstream       31 M
 libzstd-devel                          x86_64      1.5.5-1.el9                  @appstream      198 k
 llvm-devel                             x86_64      19.1.7-1.el9                 @appstream       25 M
 llvm-googletest                        x86_64      19.1.7-1.el9                 @appstream      2.3 M
 llvm-static                            x86_64      19.1.7-1.el9                 @appstream      284 M
 llvm-test                              x86_64      19.1.7-1.el9                 @appstream      2.1 M
 ncurses-c++-libs                       x86_64      6.2-10.20210508.el9_6.2      @appstream      161 k
 ncurses-devel                          x86_64      6.2-10.20210508.el9_6.2      @appstream      857 k

Transaction Summary
================================================================================================
Remove  28 Packages

Freed space: 1.1 G
Is this ok [y/N]:
```

3. Update your system libraries and packages to the latest available version in the deploymentengine session by running this command:

```
sudo yum update -y
```

```
[root@deploymentengine ~]# sudo yum update -y
Last metadata expiration check: 1:36:00 ago on Wednesday 15 October 2025 05:31:18 PM.
Dependencies resolved.
==========================================================================================
 Package                                                  Architecture
==========================================================================================
Installing:
 kernel                                                    x86_64
 kernel-core                                               x86_64
 kernel-modules                                            x86_64
 kernel-modules-core                                       x86_64
```

```
Installed:
  kernel-5.14.0-611.11.1.el9_7.x86_64              kernel-core-5.14.0-611.11.1.el9_7.x86_64
  kernel-modules-5.14.0-611.11.1.el9_7.x86_64      kernel-modules-core-5.14.0-611.11.1.el9_7.x86_64
  libwbclient-4.22.4-6.el9.x86_64                  llvm-filesystem-20.1.8-3.el9.x86_64
  openssl-fips-provider-1:3.5.1-4.el9_7.x86_64     samba-client-libs-4.22.4-6.el9.x86_64
  samba-common-4.22.4-6.el9.noarch                 samba-common-libs-4.22.4-6.el9.x86_64
  spirv-tools-libs-2025.2-1.el9.x86_64             systemtap-sdt-dtrace-5.3-3.el9.0.1.x86_64
Removed:
  kernel-5.14.0-570.33.2.el9_6.x86_64             kernel-core-5.14.0-570.33.2.el9_6.x86_64
  kernel-modules-5.14.0-570.33.2.el9_6.x86_64     kernel-modules-core-5.14.0-570.33.2.el9_6.x86_64

Complete!
[root@deploymentengine ~]#
```

Note: The yum update can take up to 15 minutes. This would normally be a scheduled task.

4. Run the following command to upgrade yum. It will only take a few seconds.

```
sudo yum upgrade -y
```

```
[root@deploymentengine ~]# sudo yum upgrade -y
PostgreSQL 18 for RHEL / Rocky / AlmaLinux 9 - x86_64
PostgreSQL 18 for RHEL / Rocky / AlmaLinux 9 - x86_64
Importing GPG key 0x08B40D20:
 Userid     : "PostgreSQL RPM Repository <pgsql-pkg-yum@lists.postgresql.org>"
 Fingerprint: D4BF 08AE 67A0 B4C7 A1DB CCD2 40BC A2B4 08B4 0D20
 From       : /etc/pki/rpm-gpg/PGDG-RPM-GPG-KEY-RHEL
PostgreSQL 18 for RHEL / Rocky / AlmaLinux 9 - x86_64
Dependencies resolved.
Nothing to do.
Complete!
[root@deploymentengine ~]#
```

5. Run the following command to clean yum:

```
sudo yum clean all
```

```
[root@deploymentengine ~]# sudo yum clean all
108 files removed
[root@deploymentengine ~]#
```

6. Run the following command as a user with sudo access to install the unzip utility.

```
sudo yum install unzip
```

```
[root@deploymentengine ~]# sudo yum install unzip
Docker CE Stable - x86_64
Extra Packages for Enterprise Linux 9 - x86_64
Extra Packages for Enterprise Linux 9 openh264 (From Cisco) - x86_64
Jenkins-stable
PostgreSQL common RPMs for RHEL / Rocky / AlmaLinux 9 - x86_64
PostgreSQL 18 for RHEL / Rocky / AlmaLinux 9 - x86_64
PostgreSQL 17 for RHEL / Rocky / AlmaLinux 9 - x86_64
PostgreSQL 16 for RHEL / Rocky / AlmaLinux 9 - x86_64
PostgreSQL 15 for RHEL / Rocky / AlmaLinux 9 - x86_64
PostgreSQL 14 for RHEL / Rocky / AlmaLinux 9 - x86_64
PostgreSQL 13 for RHEL / Rocky / AlmaLinux 9 - x86_64
Rocky Linux 9 - BaseOS
Rocky Linux 9 - AppStream
Rocky Linux 9 - Extras
Package unzip-6.0-58.el9_5.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@deploymentengine ~]#
```

Note: In our environment, the unzip utility is already installed.

7. Run the following command as a user with sudo access to set up Perl.

sudo yum install perl

```
[root@deploymentengine ~]# sudo yum install perl
Last metadata expiration check: 0:05:45 ago on Wednesday 15 October 2025 08:06:25 PM.
Package perl-4:5.32.1-481.1.el9_6.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@deploymentengine ~]#
```

Note: In our environment, Perl is already installed.

8. Run the following **two** commands as a user with sudo access to set up Perl-Data-Dumper package.

sudo yum makecache

```
[root@deploymentengine ~]# sudo yum makecache
Docker CE Stable - x86_64
Extra Packages for Enterprise Linux 9 - x86_64
Extra Packages for Enterprise Linux 9 openh264 (From Cisco) - x86_64
Jenkins-stable
PostgreSQL common RPMs for RHEL / Rocky / AlmaLinux 9 - x86_64
PostgreSQL 18 for RHEL / Rocky / AlmaLinux 9 - x86_64
PostgreSQL 17 for RHEL / Rocky / AlmaLinux 9 - x86_64
PostgreSQL 16 for RHEL / Rocky / AlmaLinux 9 - x86_64
PostgreSQL 15 for RHEL / Rocky / AlmaLinux 9 - x86_64
PostgreSQL 14 for RHEL / Rocky / AlmaLinux 9 - x86_64
PostgreSQL 13 for RHEL / Rocky / AlmaLinux 9 - x86_64
Rocky Linux 9 - BaseOS
Rocky Linux 9 - AppStream
Rocky Linux 9 - Extras
Metadata cache created.
[root@deploymentengine ~]#
```

sudo yum -y install perl-Data-Dumper

```
[root@deploymentengine ~]# sudo yum -y install perl-Data-Dumper
Last metadata expiration check: 0:02:14 ago on Wednesday 15 October 2025 08:25:21 PM.
Package perl-Data-Dumper-2.174-462.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@deploymentengine ~]#
```

Note: In our environment, Perl-Data-Dumper is already installed.

Task 2: Create Innovation Suite Namespace

For a successful installation of Innovation Suite, a namespace needs to be created. The installation will take place in next lab.

1. Create the Innovation Suite namespace:

```
kubectl create ns helixis
```

```
[root@deploymentengine ~]# kubectl create ns helixis
namespace/helixis created
[root@deploymentengine ~]#
```